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Joule Tools

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1 Using Joule with Microsoft 365 Copilot

By integrating Joule and Microsoft 365 Copilot, you can leverage the Joule capabilities from within Microsoft 365 Copilot Business Chat or vice versa. Joule is made available as a Custom Engine Agent within Microsoft 365 Copilot , and can be reached either via @-mentioning or selecting the Joule agent directly. Microsoft 365 Copilot can be reached by either providing an input in Joule that addresses Microsoft 365 Copilot's capabilities or mentioning Microsoft 365 Copilot directly.

In general, all queries that Joule can answer via the Joule Web Client are also available through Microsoft 365 Copilot. The responses from Joule are re-formatted for display within the Microsoft 365 Copilot chat. Due to the need for this transformation, there are certain content types that cannot be visualized in Microsoft 365 Copilot. In such cases, the user will receive a message indicating that the content should be accessed through the Joule Web Client instead. For more information about current limitations for Microsoft 365 Copilot with Joule, see Limitations [page 4].

> **Note**
>
> All answers from Microsoft 365 Copilot in Joule are provided in a markdown text form.

For more information about integrating Joule with Microsoft 365 Copilot, see Integrating Joule with Microsoft 365 Copilot.

> **Note**
>
> The use of Microsoft 365 Copilot is subject to the terms of the agreement between the Customer and Microsoft. SAP is not responsible for and does not control the data processed by Microsoft 365 Copilot and its compliance with any applicable laws. It is up to the customer to consider the legal implications and to decide whether to use Microsoft 365 Copilot.

Related Information

Roles and Authorizations for Joule with Microsoft 365 Copilot/Microsoft Teams [page 4]
Integrating Joule with Microsoft 365 Copilot [page 4]
Limitations [page 4]
Using Joule
Security
Accessibility Features in Joule
Monitoring and Troubleshooting
SAP Joule app

1.1 Roles and Authorizations for Joule with Microsoft 365 Copilot/Microsoft Teams

Roles and Authorizations

The user identities between Joule and connected SAP backend systems on the one hand and Microsoft 365 Copilot/Microsoft Teams on the other are linked through the corresponding trust setup between Microsoft Entra as IdP and SAP IAS as described in Integrating Joule with Microsoft 365 Copilot. For more information about administration and authentication, see User Administration and Authentication.

This means that every user question raised to Joule via the Microsoft 365 Copilot chat or Microsoft Teams chat will be processed in the scope of that user's identity and permissions. It will therefore only be answered if that user is also known to and authorized for Joule, and access will be restricted with the same permission checks that are checked when accessing Joule directly.

> **Note**
>
> There are no additional roles to be maintained for the integration.

Data, Privacy, and Security for Microsoft 365 Copilot/Microsoft Teams

For information regarding data, privacy, and security for Microsoft 365 Copilot/Microsoft Teams, see Data, Privacy, and Security for Microsoft 365 Copilot .

1.2 Integrating Joule with Microsoft 365 Copilot

To learn how to integrate Joule with Microsoft 365 Copilot, see Integrating Joule with Microsoft 365 Copilot in the Joule integration guide.

1.3 Limitations

When using Joule with Microsoft 365 Copilot the following limitations apply. There may times when Microsoft 365 Copilot directs you to your Joule Web Client for more information as a result of the limitations.

- Chat posts are limited to approximately 28 KB per post.
- Because of the chat post size limitation in Microsoft 365 Copilot, the length for lists and tables is limited so that only the first 10 items are shown. In these cases, Microsoft 365 Copilot informs the user that only the first 10 items are displayed and that more information can be accessed in Joule.
- SAP Icons are not displayed in Microsoft 365 Copilot.
- Microsoft Teams cannot show SVG images, images are limited to PNG and JPEG only.
- Adaptive Cards support only media files that have been uploaded in Share Point or One Drive.
- Adaptive Cards are limited to a maximum of six actions.
- Some advanced UI cards (for example: analytical cards, calendar cards, etc) will not be available in Microsoft 365 Copilot, as they cannot be displayed correctly in Microsoft applications. This is indicated to the user by a corresponding message, asking the user to use Joule instead.

When using Microsoft 365 Copilot with Joule the following limitations apply.

- All answers from Microsoft 365 Copilot in Joule come in a markdown text form.
- The integration uses Microsoft Chat API to display the answers. To see the current capabilities and limitations, see Copilot APIs Overview .

2 SAP Joule for Consultants

SAP Joule for Consultants is a Gen AI-powered copilot designed to boost consultant productivity by providing fast, reliable answers grounded in exclusive SAP content. It reduces rework by offering expert-level guidance that enhances design decisions and ensures alignment with best practices across project management, functional, and technical consulting. The solution draws from SAP's comprehensive knowledge base containing 18M+ pages of exclusive SAP documentation and training materials and leverages SAP's specialized fine-tuned ABAP model, trained on 300 million lines of ABAP code and 30 million lines of CDS code, to handle detailed code explanation and basic code generation. This extensive repository includes more than 9 terabytes of curated SAP knowledge content, 100+ SAP certifications, implementation guidance, product documentation, learning journeys, and insights from the SAP community.

Key Capabilities

- Enablement & Discovery
  - Provides comprehensive insights into the SAP domain by addressing questions related to product names, entity definitions, SAP concepts, programs, processes, and functional capabilities across the SAP portfolio.
- Guidance & Instructions
  - Offers step-by-step support for achieving specific objectives, including setup, configuration, customization, and transformation of SAP systems and business processes across the SAP product suite.
- Support & Troubleshooting
  - Delivers precise solutions for resolving functional issues or operational challenges within SAP systems and business processes across the SAP ecosystem.
- ABAP Code Interpretation
  - Deciphers ABAP code by explaining its business logic and technical characteristics.

> **Note**
>
> Please refrain from entering any personal data into the prompt field, as it is not intended or designed to process, store, or handle personal information.

Getting Started with SAP Joule for Consultants

- Access SAP Joule for Consultants
  - Use the dedicated URL provided by your organization to access the SAP Joule for Consultants landing page. For more information about using Joule as a standalone application, see Using the Joule Web Client.
- Ensure your account permissions and registration are set up by your admin.
- Landing Page Overview
  - You'll see a welcoming message: "Hello, Welcome to SAP Joule for Consultants!"
- Start a conversation by typing your question in the input field.

Using SAP Joule for Consultants

- Ask Questions
  - Enter clear and specific queries like, " What are the key deliverables during the Discovery phase according to the RISE Methodology?" to get context-rich answers.
- SAP Joule for Consultants uses SAP's knowledge base to provide concise, accurate responses with source references for deeper insights.
- Conversation Settings
  - Conversations expire after 8 hours of inactivity. You'll be notified and can start a new conversation.
- Feedback (Optional)
  - Use thumbs-up or thumbs-down to rate responses. Feedback helps improve SAP Joule for Consultants and is anonymous.

Example Queries:

- How can I optimize freight costs using advanced algorithms and real-time data in SAP Transportation Management (TM), while ensuring carrier performance monitoring and collaboration with external partners?
- Generate ABAP code for a sales price calculation BAdI that decreases the location specific price by 10% per for every 2 degrees above 20 Celsius.

Related Information

Using the Joule Web Client
Security
Accessibility Features in Joule
Monitoring and Troubleshooting

2.1 Provisioning SAP Joule for Consultants

Prerequisites

You must have, or be able to access and create the following:

- A Global BTP Account
- SAP Cloud Identity Services
- Access to EU 10 or EU 11 or US 10 region
- SAP AI Units

> **Note**
>
> SAP AI Units are a commercial metric used to track and manage the consumption of AI-related services within the SAP Business Technology Platform. They work like credits or tokens and are required to access services such as SAP AI Core, SAP Joule, AI Foundation, and other AI-powered or machine learning functionalities.
>
> A new SKU – SAP Joule for Consultants (SKU: 8019164) is now available and must be purchased. Any SAP salesperson can license the SAP AI Unit SKU. Please ensure that you have signed the applicable Terms & Conditions (T&Cs) with your Account Executive (AE) before proceeding.

BTP Global Account

SAP Joule for Consultants is a SaaS application built on SAP's Business Technology Platform (BTP). Once commercial agreements are completed, an entitlement to SAP Joule for Consultants is assigned to your BTP global account, enabling you to provision and access the service.

A global account in SAP BTP represents the contractual agreement between a customer or partner and SAP. It serves as the foundation for managing subaccounts, users, entitlements and quotas.

- If you already have a BTP global account, you can proceed to provision SAP Joule for Consultants in the dedicated subaccount.
- If you do not yet have a global account, please follow the steps to create one before provisioning the service below.

Create a BTP Global Account

To create a BTP global account, follow these steps:

1. Receive the welcome email from SAP (no.reply.provisioning@sap.com) sent to the IT Contact person of your company.
2. Look for a second email containing your logon information, including an S-user ID.
3. Use the link provided in the email or go directly to the SAP BTP Cockpit Home (https://account.hana.ondemand.com/#/home/welcome).
4. Sign in using the global account administrator credentials received in your email. Once logged in, you will see your global account(s) in the SAP BTP Cockpit Home. After creating your BTP global account, it's recommended to:
5. Add at least one additional global account administrator for support and backup.
6. Change the display name of your global account if desired.

Check your entitlements for the global account to understand your available resources and services.

Remember, the initial access is provided only to the IT Contact, who is responsible for adding additional global administrators and delegating work as required.

For a step-by-step guide on how to get a BTP global account refer to this learning journey.

Enable SAP Joule for Consultants in your SAP BTP Account

Enabling SAP Joule for Consultants in your BTP global account involves checking service entitlements, configuring trust with IAS, creating and verifying service instances, launching the Joule Booster, activating the service via SAP for Me, managing user access in IAS, and finally launching the SAP Joule UI.

1. **Check the Service Entitlement in the BTP Cockpit**
   - Go to your global account in the BTP Cockpit
   - Navigate to Entitlements > Service Assignments
   - Check entitlements:
     - Joule
     - SAP Joule for Consultants
2. **BTP subaccount**
   - Create a subaccount and select region EU 10 or EU 11
   - Add the service plans
     - Joule with plan foundation
     - SAP Consulting Capability with plan standard-ga
3. **Configure Trust with Identity Authentication (IAS)**
   - Go to Security → Trust Configuration
   - Select the appropriate IAS tenant and establish trust

   For more information, see Configure Trust to the Identity Authentication Tenant
4. **Subscribe to SAP Joule for Consultants**
   - Navigate to Instances and Subscriptions and click *Create*
   - Select the service SAP Consulting Capability and plan standard-ga
   - Follow the prompts and deploy the instance
   - Verify Instance Creation and Confirm System Landscape
     - In the global account, go to System Landscape > Systems
     - Ensure that SAP Consulting Capability is now displayed.
5. **Run the Joule Booster**
   - In the global account, run the booster **Setting Up Joule**
   - During configuration, select only SAP Consulting Capability as the product

   > **Note**
   >
   > *SAP Consulting Capability cannot be included in the same Joule Tenant with other SAP products. Please use a different sub-account and create a new formation for integrating your other SAP products with Joule.*

   For more information, see Run the Joule Booster
6. **Role Assignment**
   - In the sub account, open Security > Role Collections
   - Create a role collection
   - Assign the role `end_user` from the application das-application!b188376
   - Add a user group with name SAP_Joule_END_USER
7. **Access SAP for Me and Request Activation**

Create a ticket in SAP for Me and include the following information:

- Support component: CA-JOULE-BAI
- URL of the SAP Cloud Identity Services tenant
- PUPM package: SAP Joule for Consultants
- Subaccount ID

8. **Open SAP Cloud Identity Services and Manage User Group**

> **Note**
>
> **Ensure the ticket has been processed from the previous step before completing this step.**

- Open SAP Cloud Identity Services
- Navigate to *Users & Authorizations* and open *Groups*
- Search for the SAP_Joule_PREMIUM_CONSULTANT
- Add the users who should have access to SAP Joule for Consultants
- Create a User Group with name `SAP_Joule_END_USER`
- Add the same users

9. **Launch SAP Joule for Consultants**
   - Go back to BTP and select your Joule instance
   - Login using your IAS with your credentials
   - The screen should be displayed with the message, **The service is up and running**.
   - Once logged in, append **/Joule** to the URL and press **Enter**
     You should see the SAP Joule for Consultants UI load successfully

> **Note**
>
> If you encounter any issues during provisioning or with Joule for Consultants, refer to the Monitoring and Troubleshooting guide for support.

Related Information

Configure Trust to the Identity Authentication Tenant
Run the Joule Booster
Managing Entitlements and Quotas Using the Cockpit

2.2 SAP Joule for Consultants Console

The SAP Joule for Consultants Console provides access to up to date release notes, usage metrics, an integrated **Prompt Library**, and system settings for both Standard and Administration-level users.

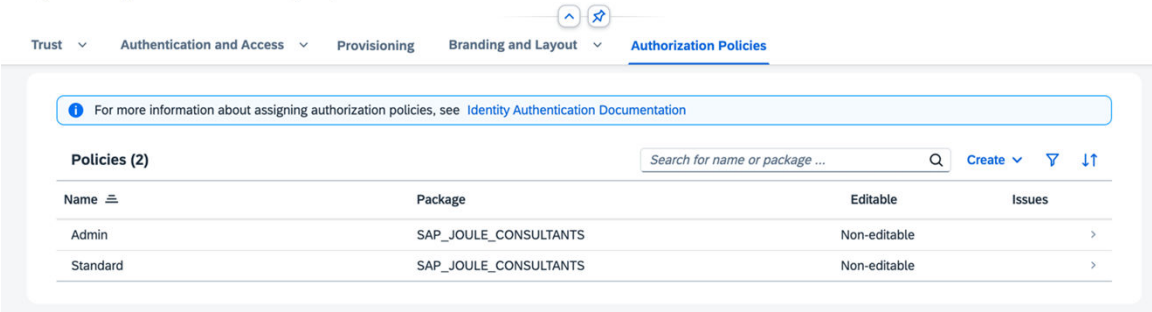
Accessing the SAP Joule for Consultants Console

1. Log in to your SAP BTP subaccount.
2. In the subaccount, click on the Joule for Consultants subscription link.

> **Note**
>
> The Joule for Consultants Console uses two built-in access role policies:
>
> - Admin – Complete access to all console features.
> - Standard – Limited access. (Currently optional.)
>
> Anyone in your IAS tenant can open the console, but only Admins have full access.

Assigning Admin Rights

1. In IAS, go to *Application & Resources* > *Applications*.
2. Search for scc-identity-service and choose the application for your subaccount.
3. Open Authorization Policies. Both Admin and Standard roles are displayed.



4. To assign Admin rights:
   - Go to *Users & Authorizations* > *Groups*.
   - Select SAP_JOULE_CONSULTANTS – Admin.
   - Add the users who need full Admin privileges.

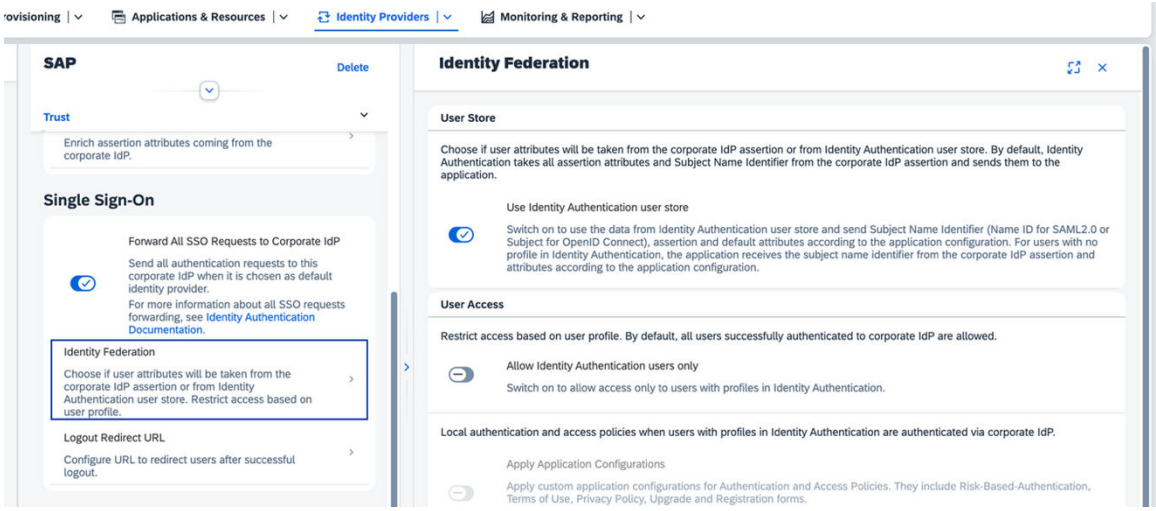
> **Note**
>
> Double-check the **Application Name** to make sure you are adding users under the correct subaccount.

Using Corporate Identity Providers (IDP)

If your organization uses a corporate IDP, you must align the settings for both Joule and Joule for Consultants.

1. In *Identity Providers* > *Corporate Identity Providers*, select your corporate provider.
2. Make sure the toggle *Use Identity Authentication user store* is enabled.

> ⚠ Caution
>
> If the toggle, *Use Identity Authentication user store* is not enabled, group authorizations will not work.



> → Remember
>
> The user must already exist in the IAS user store before you can add them to an authorization group.

3 Joule Analytics Center

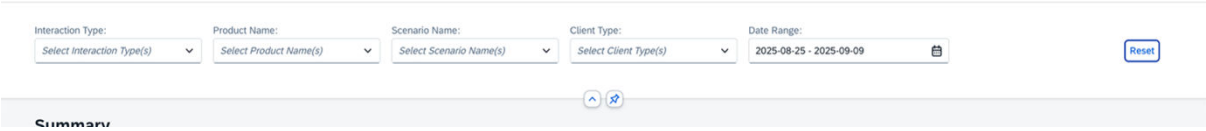
Introduction

Joule Analytics Center is a tenant specific dashboard that displays **Product Usage**, **Scenario Usage**, **Interaction Type Usage** and **Client Type Usage**.

Using Joule Analytics Center

Filters

Joule Analytics Center provides five filters located at the top of the page. You can use the filters to narrow down the data you want to view by selecting as many options as you like. To clear filter selections, you using the **Reset** button on the right.



Available filters:

- **Interaction Type**: The type of interaction the user has with Joule. There are eight main types:
  - Direct response
    - When Joule's response contains generic answers.
    - For example, when you say *Bye*, and Joule responds with *Bye, have a nice day*.
  - Scenario
    - When Joule returns business data of the corresponding prompt or the app navigation button for accessing that data information. This means the corresponding SAP Products capability scenario was selected and executed by Joule.
    - For example, when asking *Who are my colleagues* returns the data for your colleagues.
  - Information retrieval
    - This contains the documents source links from the SAP Help Documents or the Custom Documents uploaded into the customer's *document grounding* repository.
  - Slot filling
    - When Joule asks for more information from the customer before directing the request to a scenario.
    - For example, if you type *I want to update my email address*, Joule will respond with a follow-up question of *What do you want to update your email address to?*. This response is known as slot filling.
  - File upload
    - When the user uploads a file to Joule.
  - Disambiguation

    - When `Joule` returns multiple capability scenario options for you to choose from. You can then select a scenario that best fits your needs.
  - File Q&A
    - When the user asks questions about a previously uploaded (see File Upload) or existing document.
  - Web Search
    - When the user request requires external information, search results from the web in general.
    - For example, if you type *What are the chances the Leafs make the playoffs this year?* and `Joule` responses with an answer that uses information available on the Internet.
- **Product Name**: The name of the product for which `Joule` is used. For example, *SuccessFactors*.
- **Scenario Name**: The scenario under which `Joule` is used. These names are designed by LOBs during product development.

  > **Note**
  >
  > Only messages or conversations with scenario, slot filling interaction type have a Scenario Name.

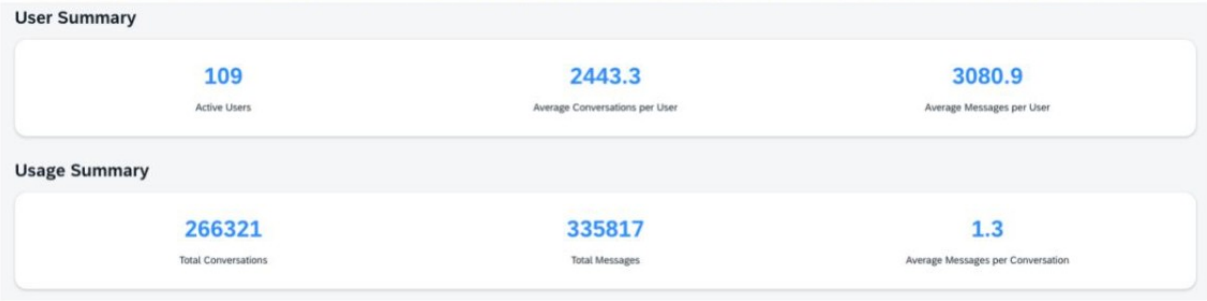
- **Client Type**: Where `Joule` is used, for example, *desktop*, *mobile* or *copilot*.
- **Date Range**: The date range of the data you are interested. Minimum date is the date JAC is released, and maximum date you can select is the current date. You will not be able to include future dates in your date range selection.

Charts

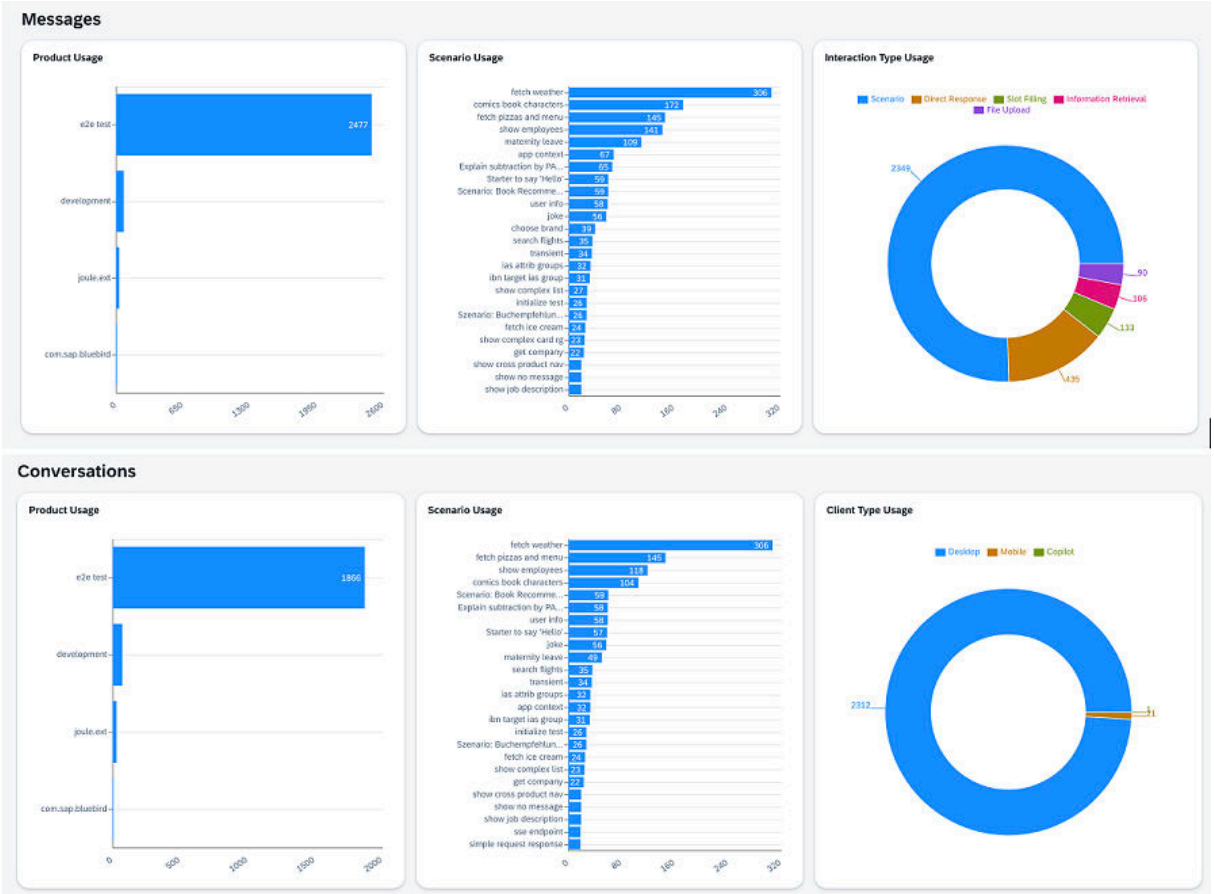
The `Joule` Analytics Center dashboard includes both summary and charts.

The **User Summary** panel shows an overview of user data, including the number of unique active users, average conversations per user, and average messages per user.

The **Usage Summary** panel shows an overview of usage data, including total number of conversations, total number of messages and average messages per conversation.

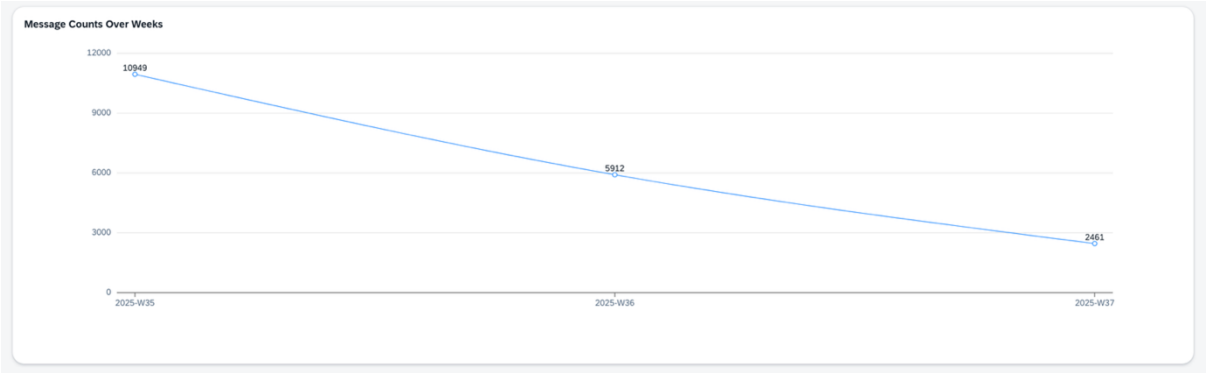


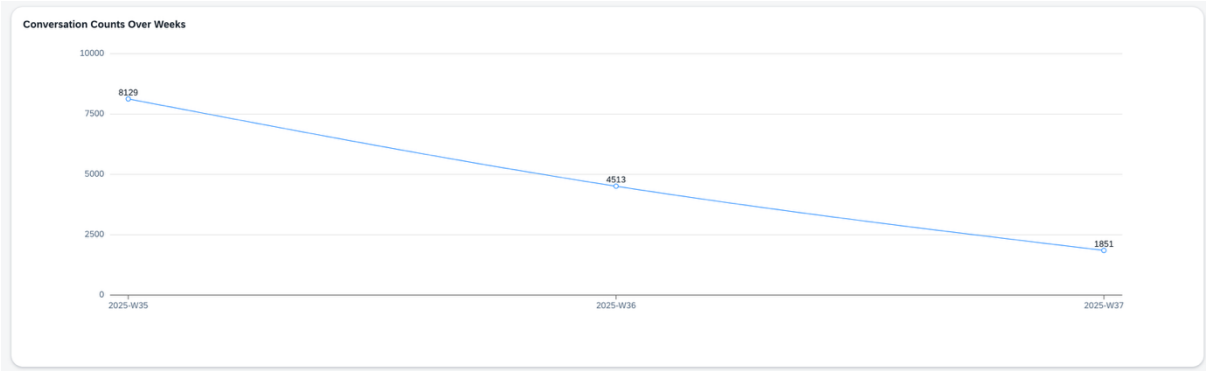
The **Messages** and **Conversations** section shows a more detailed breakdown of the number messages or conversations as per Product Usage, Scenario Usage, Interaction Type Usage (for messages), and Client Type Usage (for conversations).



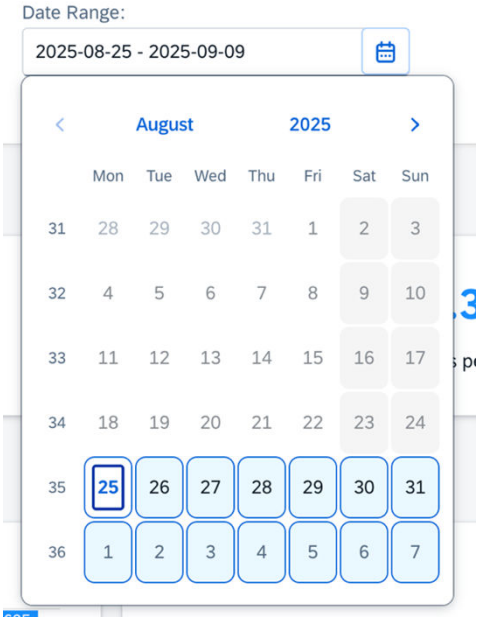
If you are interested in a particular product, scenario name, or interaction type, click on the **bar** you are interested, and the filter will automatically update to show only that product, scenario name, or interaction type.

The line charts for Message Counts Over Weeks and Conversation Counts Over Weeks shows the trend of `Joule` usage over time.





In these examples, *2025-W35* means the 35th week of 2025. To check which dates the week refers to, you can click open the **Date Range filter** and refer to the week number labelled at the start of each week. Week 35 refers to the week of 25 Aug – 31 Aug.



Related Information

Enabling Joule Analytics Center [page 17]

3.1 Enabling Joule Analytics Center

Prerequisites

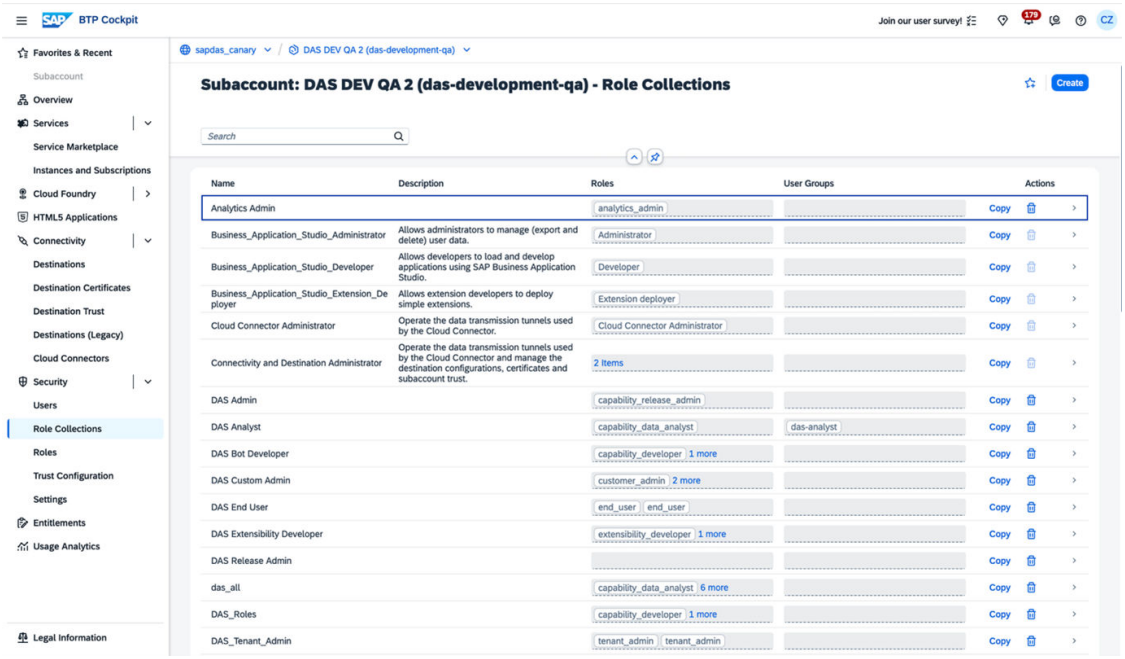
- You have an existing SAP BTP account with a Joule Subscription
  For more information, see https://help.sap.com/docs/joule/integrating-joule-with-sap/prerequisites?locale=en-US.
- All users requiring access to Joule Analytics Center have IAS configured
  For more information, see https://help.sap.com/docs/joule/integrating-joule-with-sap/configure-trust-to-identity-authentication-tenant?locale=en-US&q=IAS.

Accessing Joule Analytics Center

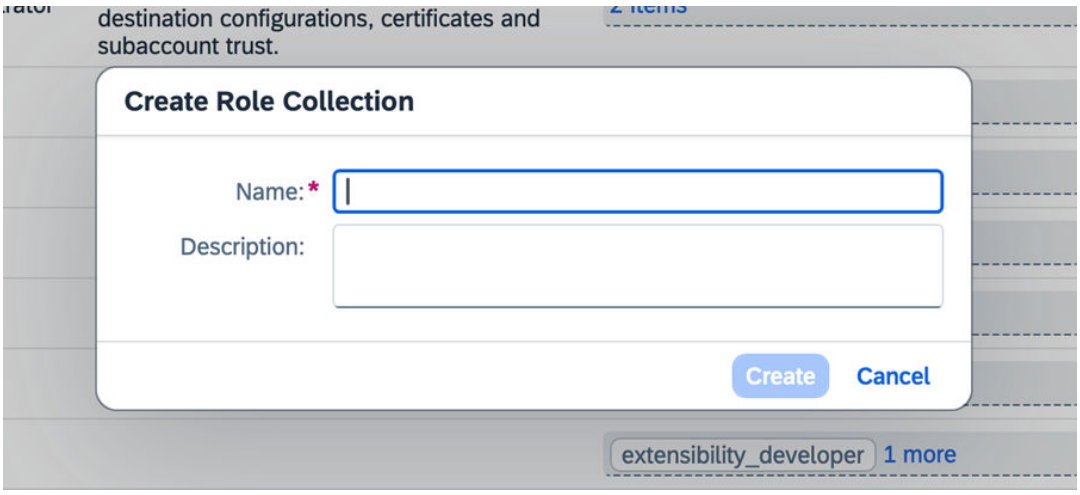
1. Create an *Analytics Admin* role collection.

   > **Note**
   >
   > This step only needs to be done once.

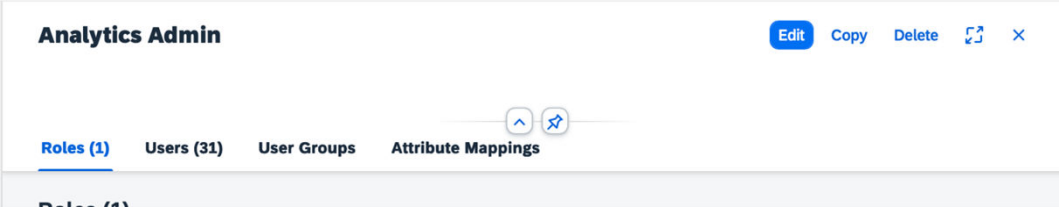
   1. If you are an administrator of the subaccount, go to `Security > Role Collections` on the SAP BTP Cockpit:



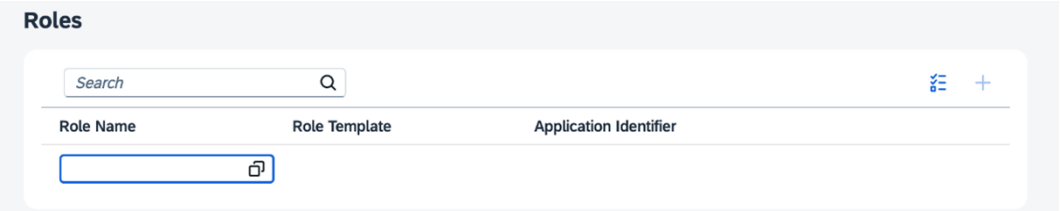
1. If you do not see an Analytics Admin role collection, click the **Create** button on the top right corner of the page.

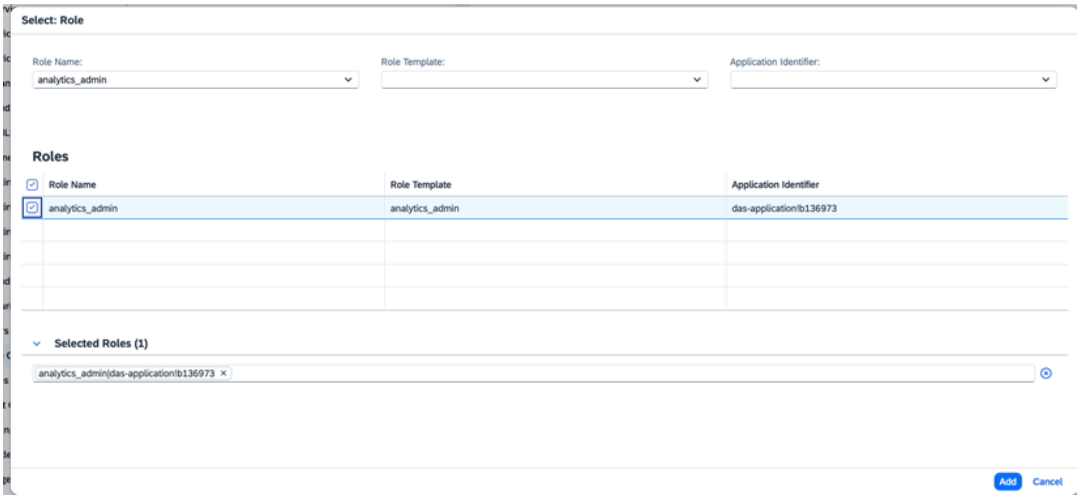


2. Enter *Analytics Admin* (or a collection name of your choice) in the *Name* field and click **Create**.
3. Click on the role collection you just created and click **Edit**.

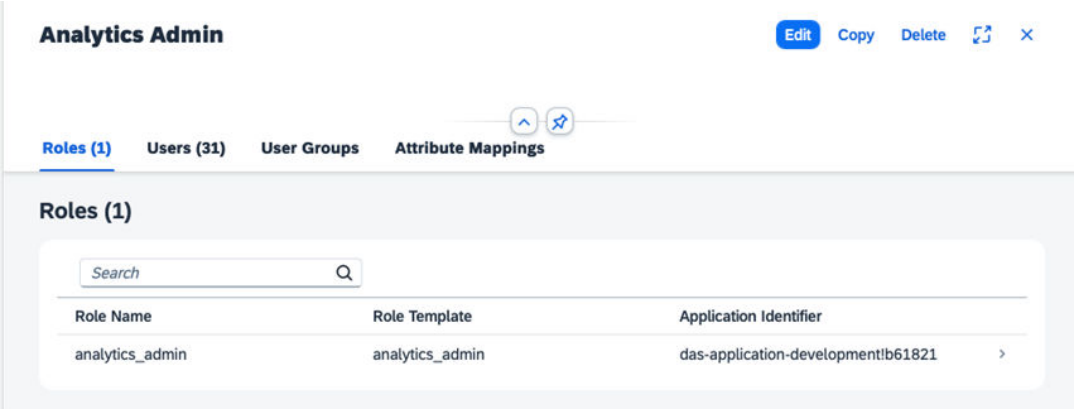


4. Under *Roles*, look for *analytics_admin* role, check the checkbox and click **Add**.



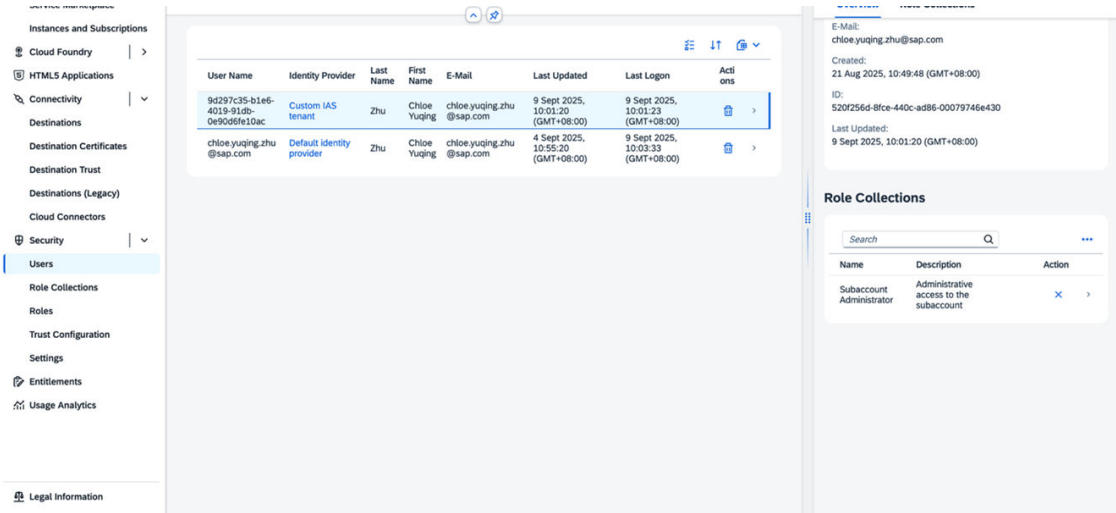


5. You should see the role added to the role collection.

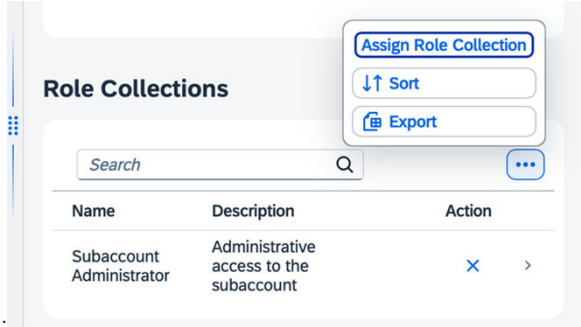


> **Note**
>
> If you are not an administrator of the subaccount, contact your administrator to complete the above steps.

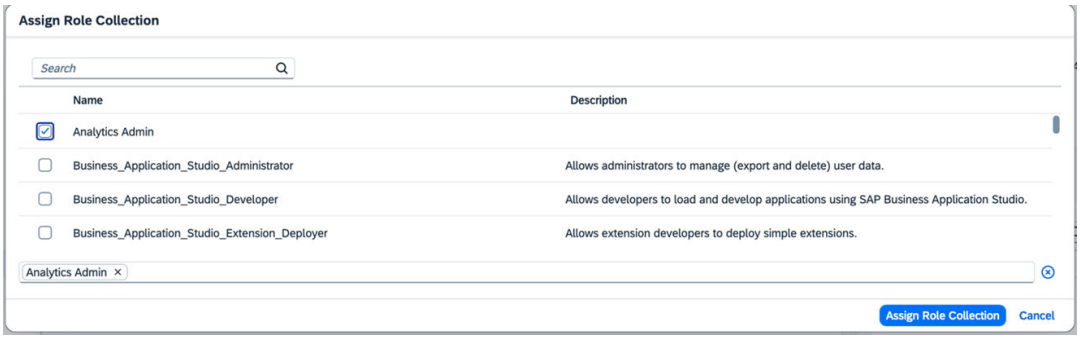
2. All users must have the role of *Analytics Admin*.
   1. If you are administrator of the subaccount, you can grant the role to other users under `Security > Users` on the SAP BTP Cockpit:

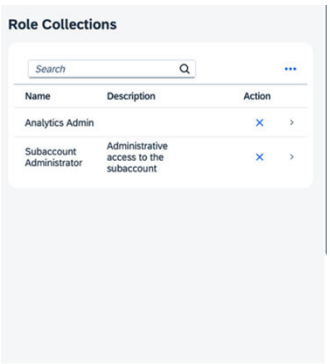


   1. Look for the user you want to grant the role to, under *Role Collections*, click on the **three dots**, and then click **Assign Role Collection**.



2. Tick the box for *Analytics Admin* and then click the **Assign Role Collection** button.

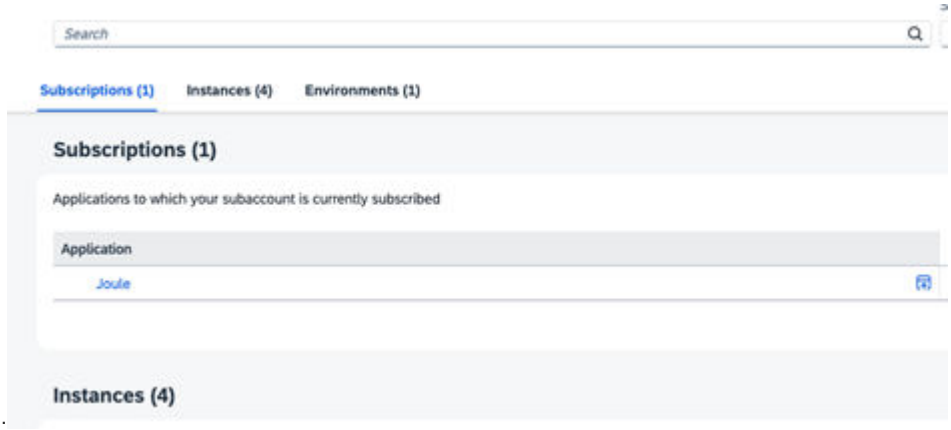




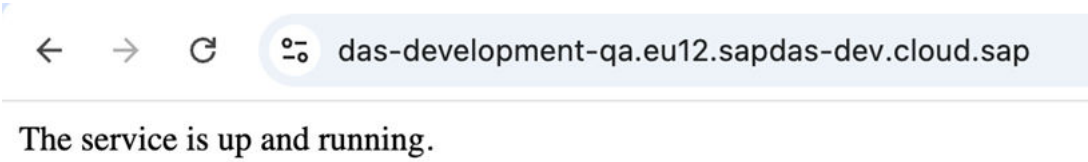
3. The assigned user now has the *Analytics Admin* role.

> **Note**
>
> If you are not an administrator of the subaccount, contact your administrator to complete the above steps.

3. On the SAP BTP Cockpit, under *Instances and Subscriptions*, look for *Joule instance* under *Subscriptions*.

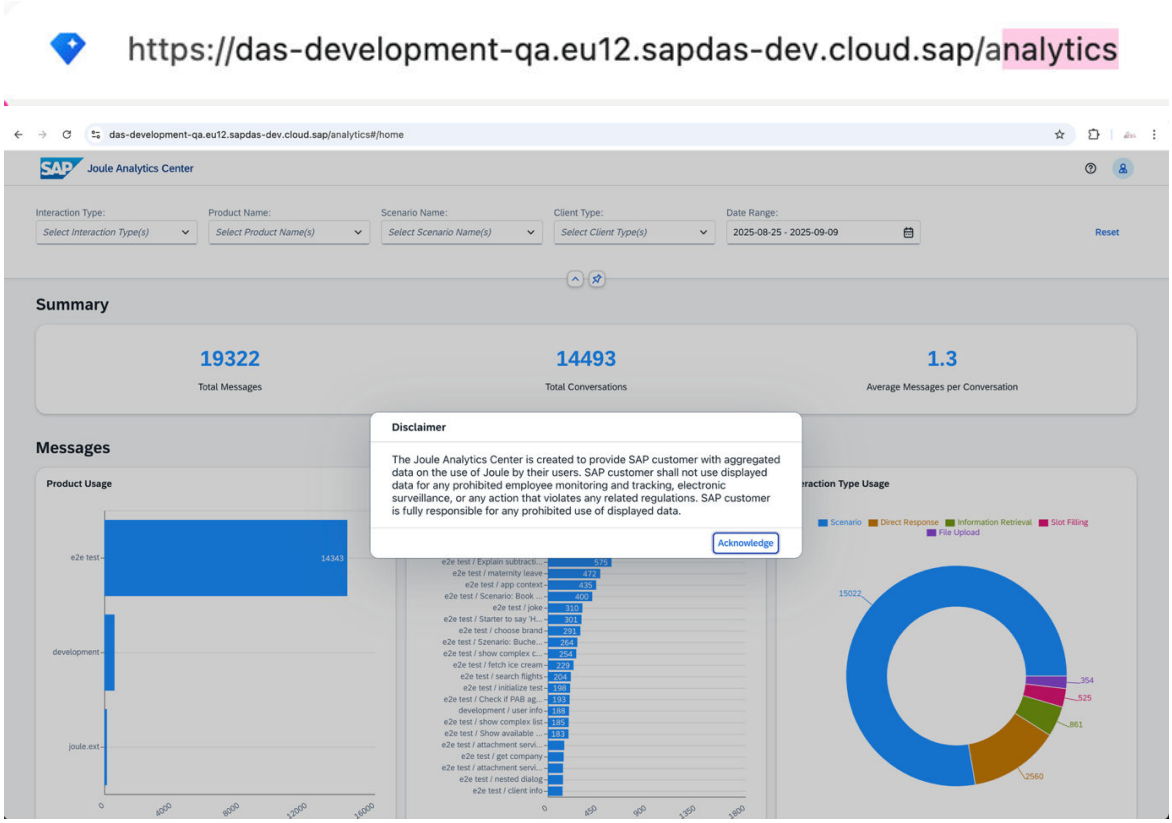


4. Click on the **Joule instance**. The following page will open.



this:

5. Append */analytics* to the URL, and you will be redirected to the Joule Analytics Center page.



Processing of Personal Data for the Provision of Joule Analytics Center

For the provision of the Joule Analytics Center, we do not store direct identifiers such as email, I-number or UUID, but only pseudonymized data that cannot be reversed to an individual without access to the original identifier. For more information, seeData Protection and Privacy.

3.2 Limitations

Joule Analytics Center Known Limitations

- Joule Analytics Center will only show data when there are 8 or more individual users using Joule. This is due to legal requirements and to prevent employee monitoring and tracking or electronic surveillance.
- For interaction types of **Direct Response**, **Information Retrieval**, **Disambiguation**, **File Q&A**, **Web Search**, there is no associated product or scenario. In this case, there is no data for the related charts when you filter by **Direct Response**, **Information Retrieval**, **Disambiguation**, **File Q&A** or **Web Search**.
- Joule Analytics Center only provides data beginning from the Joule Analytics Center release date. Data previous to this date cannot be retrieved.

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